

cand_003 redocked against MAO-A, MAO-B, TTBK1, TTBK2

Clean-receptor protocol (no water, no heteroatoms, missing atoms rebuilt, polar H, Gasteiger charges) · AutoDock Vina 1.2.7 · known active site, exhaustiveness 32 · 3 seeds

Result

Receptor	Target	Best score (kcal/mol)	Seeds 11/22/33	Pose vs crystal site	Verdict
2V5Z	MAO-B	-11.15	-11.15/-11.14/-11.14	0.8 A from native SAG centroid	Reliable: in the substrate cavity, no clash with FAD or waters
2Z5X	MAO-A	-8.77 (raw top mode -11.44 is an artefact)	-11.38/-11.41/-11.44 raw; clash-free mode 6 = -8.77 in all 3 seeds	raw pose 3.5 A off native and inside FAD	Use -8.77. Raw modes 1-5 overlap the deleted FAD
7JXX	TTBK1	-8.49	-8.49/-8.48/-8.49	1.5 A from native VP7 centroid	Reliable, in the ATP site
7Q8Y	TTBK2	-9.87	-9.86/-9.87/-9.86	3.5 A from native 9IV centroid	Treat with caution: pose is off the crystal binding mode (see notes)

Scores are Vina mode-1 affinities of the best of three seeds; the seeds agree within 0.03 kcal/mol for MAO-B, TTBK1 and TTBK2 and within 0.06 for the raw MAO-A top mode, so the search is converged and the differences between receptors are not sampling noise.

Two things you must know before using these numbers

1. MAO-A: the top score is an artefact of removing FAD. Modes 1-5 of every seed place 4-12 ligand heavy atoms within 2.5 A of where FAD sits in the crystal (closest approach 0.3-0.55 A); the CF3 end of the ligand lies inside the flavin ring system (see figure). With the cofactor deleted, nothing penalised that. The best mode that avoids FAD is **-8.77 kcal/mol**, and the project's earlier MAO-A run that kept FAD (no waters) gave -8.19 to -8.24, so -8.77 is the number to quote. Both poses are supplied.

2. Stripping every heteroatom is not neutral for these targets. In the crystals FAD lines the MAO cavity and TTBK2's 9IV inhibitor binds through a bridging water. In the same frame, 17 of the ligand's atoms in the TTBK2 pose overlap crystal waters (0 for MAO-B, 3 within 3.5 A for TTBK1), which is consistent with the pose settling 3.5 A away from the crystal ligand. The project's own dry-vs-water test found the same TTBK2 sensitivity. I followed your instruction as given; if you want a like-for-like comparison, the receptors with FAD and the pocket-water shell are already in 03_receptors/ and can be docked the same way.

Comparison with the project's earlier cand_003 runs (Vina, best of 3 seeds)

Receptor	This run (all heteroatoms removed)	Earlier: no water, cofactor kept	Earlier: pocket waters + cofactor kept
MAO-B 2V5Z	-11.15	-11.42	-12.10
MAO-A 2Z5X	-11.44 raw / -8.77 clash-free	-8.24	not run for candidates
TTBK1 7JXX	-8.49	-8.41	-9.22
TTBK2 7Q8Y	-9.87	-9.89	not run (needs bridging-water variant)

Removing waters costs roughly 0.7-0.8 kcal/mol on MAO-B and TTBK1, as expected. Only MAO-A moves in the other direction, and that is the FAD artefact above.

Preparation and docking protocol (your steps 1-8)

Step	What was done
1. Remove water	All HOH removed (2V5Z 717, 2Z5X 182, 7JXX 291, 7Q8Y 739 in the raw file). Chain A only, first alternate location.
2. Remove heteroatoms	2V5Z: FAD, SAG (native). 2Z5X: FAD, HRM (native), DCX, GOL. 7JXX: VP7 (native), Na. 7Q8Y: 9IV (native), PO4.
3. Missing atoms	PDBFixer 1.x: 3 heavy atoms rebuilt in 2V5Z, 0 in 2Z5X, 38 in 7JXX, 4 in 7Q8Y (+ C-terminal OXT). Unresolved residues were not built (invented loop coordinates): 2V5Z N-term MET-SER and 19-residue C-terminal tail; 7JXX N-term 10 residues, 1 residue after 22, 2 after 164, and a 31-residue C-terminal segment; 7Q8Y N-term 6 residues. 2Z5X is complete. The 7JXX gaps at 22 and 164 are chain breaks.
4. Polar hydrogens	Hydrogens added at pH 7.4 (PDBFixer), then written to PDBQT by Meeko with non-polar H merged and polar H typed HD. Meeko was used rather than MGLTools, because MGLTools failed to bond PDBFixer's hydroxyl H (Ser/Thr/Tyr) and gave several Asn/Arg residues wrong charges.
5. Charges	Gasteiger. Verified: every residue's charge is a whole number, and totals match the protonation states exactly: 2V5Z -3, 2Z5X +2, 7JXX +12, 7Q8Y +11. Only chain termini are fractional. Ligand cand_003 sums to -0.001 (neutral, 26 atoms in the PDBQT, 2 polar H).
6. Site and exhaustiveness	Active site is known, so this is not blind docking. Box = the project's native-ligand box (18 Å cube; 18 x 18.9 x 18 for 2V5Z), centred on the crystal ligand. Exhaustiveness 32 (not 100), 9 modes, seeds 11/22/33.
7. Results	This report, all_modes_scores.csv (108 rows), and per-receptor complex PDBs.
8. Software	AutoDock Vina 1.2.7, Vina scoring function, on AutoDock-format PDBQT files. AutoDock4 (ad4) was not used.

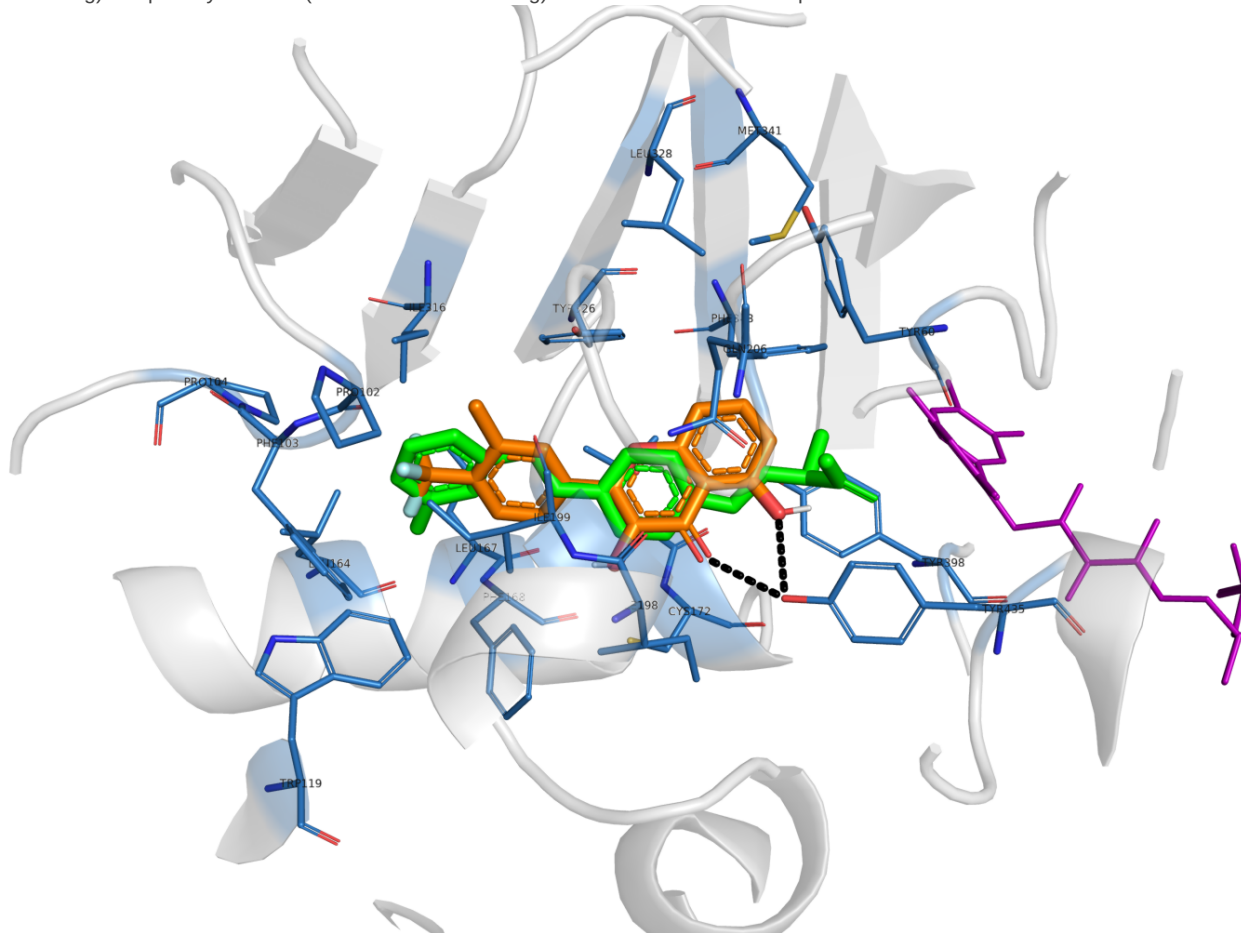
Files for Discovery Studio

Discovery Studio does not open PDF files as structures, so the structures are supplied as **PDB**:

*_best_pose_complex.pdb (protein with hydrogens + docked ligand, residue C03, chain L, with full CONECT bonding) opens directly; *_all9_poses.pdb steps through all nine modes as MODELS; *_best_pose_ligand.pdb / .mol are the ligand alone. This PDF is the summary and figures. Ligand hydrogens are explicit; protein hydrogens are those added at pH 7.4.

cand_003 in MAO-B (2V5Z)

Vina score **-11.15 (seed 11, mode 1) kcal/mol**. Orange: cand_003. Blue: receptor residues within 4 Å. Green: crystal ligand (removed before docking). Purple: crystal FAD (removed before docking). Black dashes: N/O...N/O pairs within 3.3 Å.



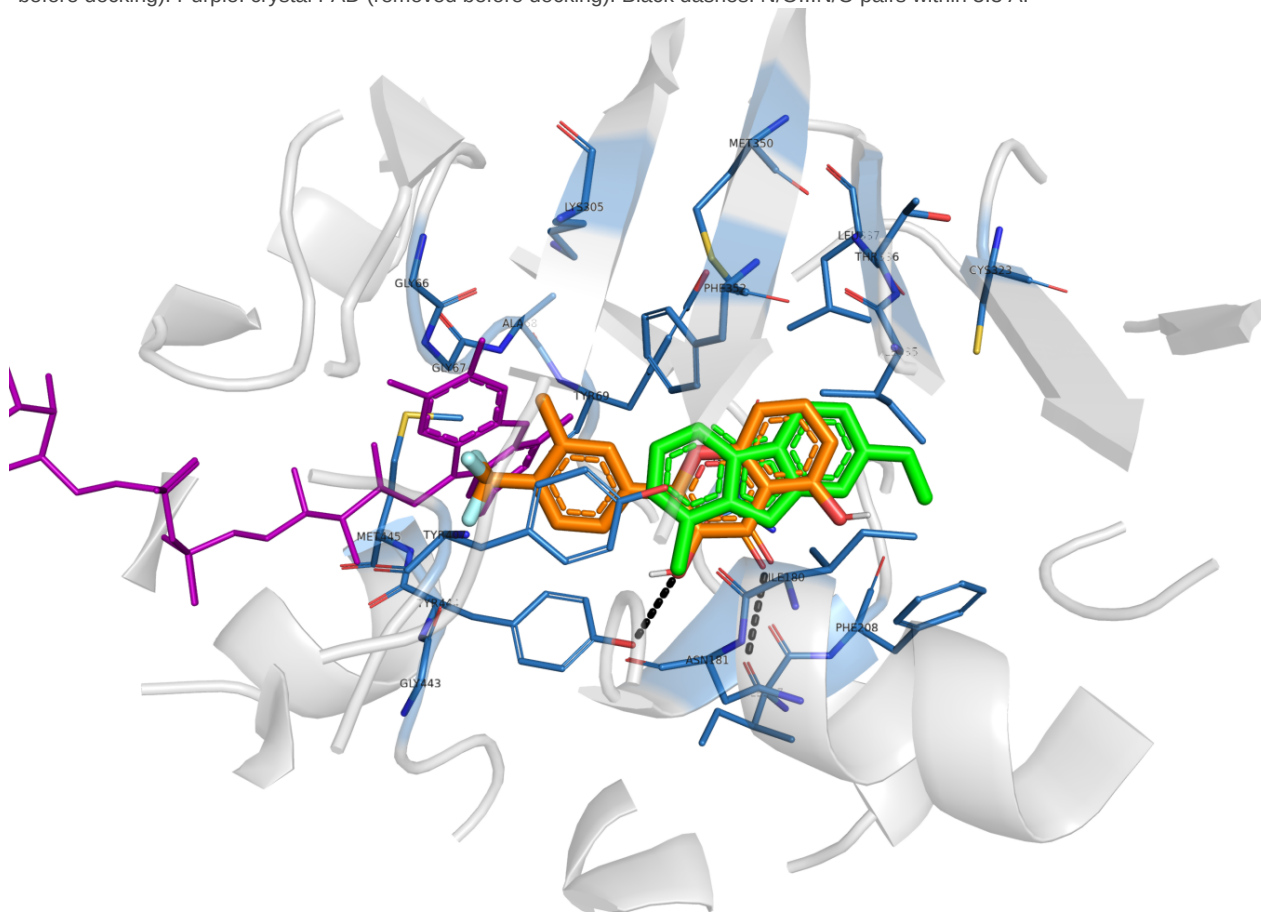
Residues within 4 Å: PRO102 TRP119 LEU164 PHE168 LEU171 CYS172 ILE198 ILE199 GLN206 ILE316 TYR326 PHE343 TYR435

Hydrogen-bond candidates (heavy-atom N/O...N/O ≤ 3.3 Å): ILE198:O ... lig O17 (3.17 Å); TYR435:OH ... lig O12 (3.10 Å); TYR435:OH ... lig O15 (3.02 Å)

Distance to removed crystal heteroatoms (raw-pose): FAD: closest 4.68 Å; waters: 3 ligand atoms within 3.5 Å.

cand_003 in MAO-A (2Z5X), raw top mode, overlaps FAD

Vina score **-11.44 (seed 33, mode 1) kcal/mol**. Orange: cand_003. Blue: receptor residues within 4 Å. Green: crystal ligand (removed before docking). Purple: crystal FAD (removed before docking). Black dashes: N/O...N/O pairs within 3.3 Å.



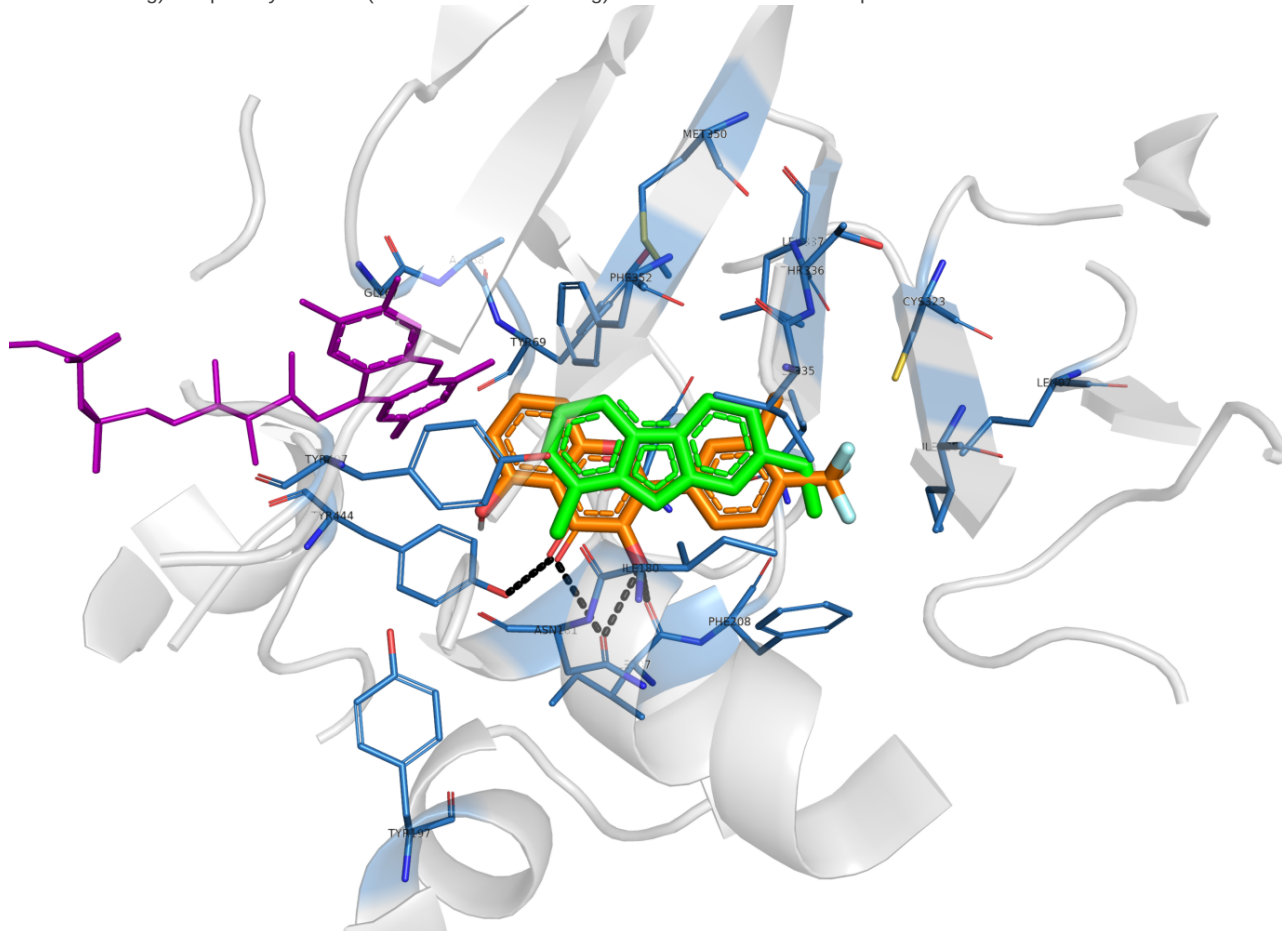
Residues within 4 Å: GLY67 ALA68 ILE180 ASN181 ILE207 PHE208 GLN215 ILE335 LEU337 MET350 PHE352 TYR407 GLY443 TYR444

Hydrogen-bond candidates (heavy-atom N/O...N/O ≤ 3.3 Å): ASN181:OD1 ... lig O15 (3.18 Å); TYR444:OH ... lig O17 (3.25 Å)

Distance to removed crystal heteroatoms (raw-pose): FAD: closest 0.53 Å; DCX: closest 18.24 Å; GOL: closest 9.81 Å; waters: 14 ligand atoms within 3.5 Å.

cand_003 in MAO-A (2Z5X), best FAD-clash-free pose

Vina score **-8.77 (seed 11, mode 6) kcal/mol**. Orange: cand_003. Blue: receptor residues within 4 Å. Green: crystal ligand (removed before docking). Purple: crystal FAD (removed before docking). Black dashes: N/O...N/O pairs within 3.3 Å.



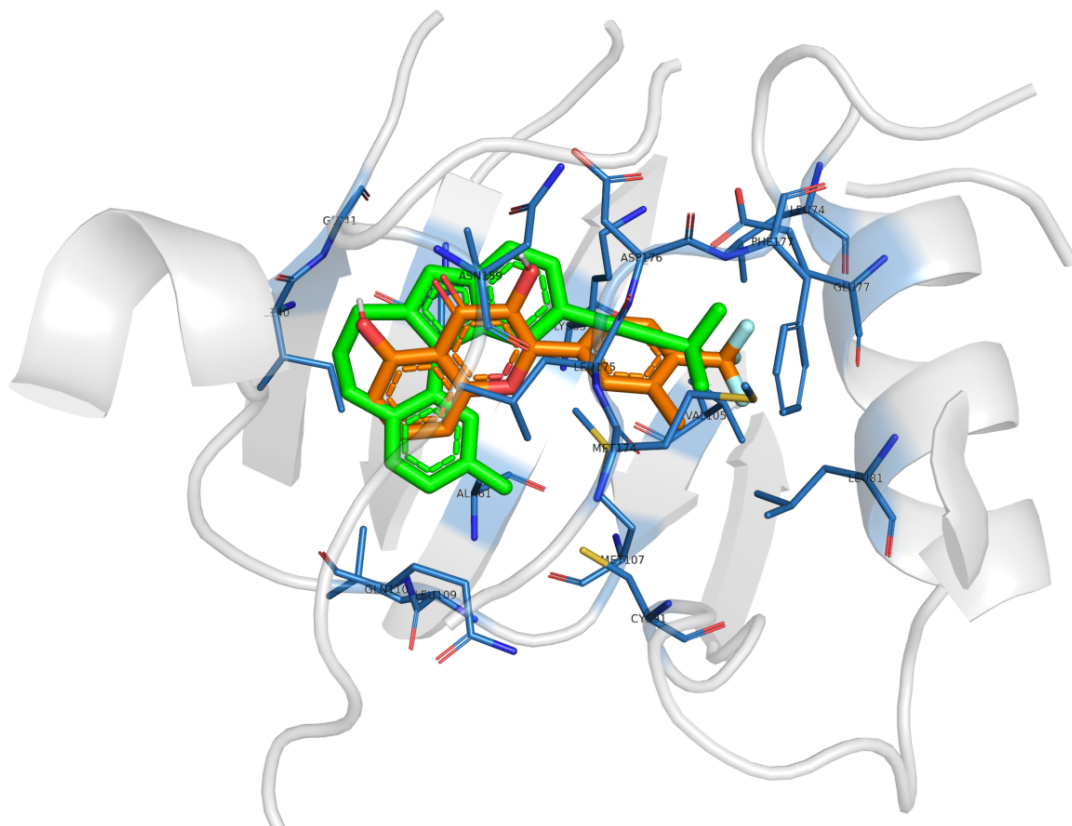
Residues within 4 Å: TYR69 ILE180 ASN181 ILE207 PHE208 VAL210 GLN215 CYS323 ILE325 ILE335 THR336 LEU337 PHE352 TYR407 TYR444

Hydrogen-bond candidates (heavy-atom N/O...N/O ≤ 3.3 Å): ASN181:OD1 ... lig O15 (3.03 Å); ASN181:OD1 ... lig O17 (2.63 Å); ILE207:O ... lig O17 (3.08 Å); TYR444:OH ... lig O15 (2.77 Å)

Distance to removed crystal heteroatoms (raw-pose): FAD: closest 0.53 Å; DCX: closest 18.24 Å; GOL: closest 9.81 Å; waters: 14 ligand atoms within 3.5 Å.

cand_003 in TTBK1 (7JXX)

Vina score **-8.49 (seed 11, mode 1) kcal/mol**. Orange: cand_003. Blue: receptor residues within 4 Å. Green: crystal ligand (removed before docking). Purple: crystal FAD (removed before docking). Black dashes: N/O...N/O pairs within 3.3 Å.



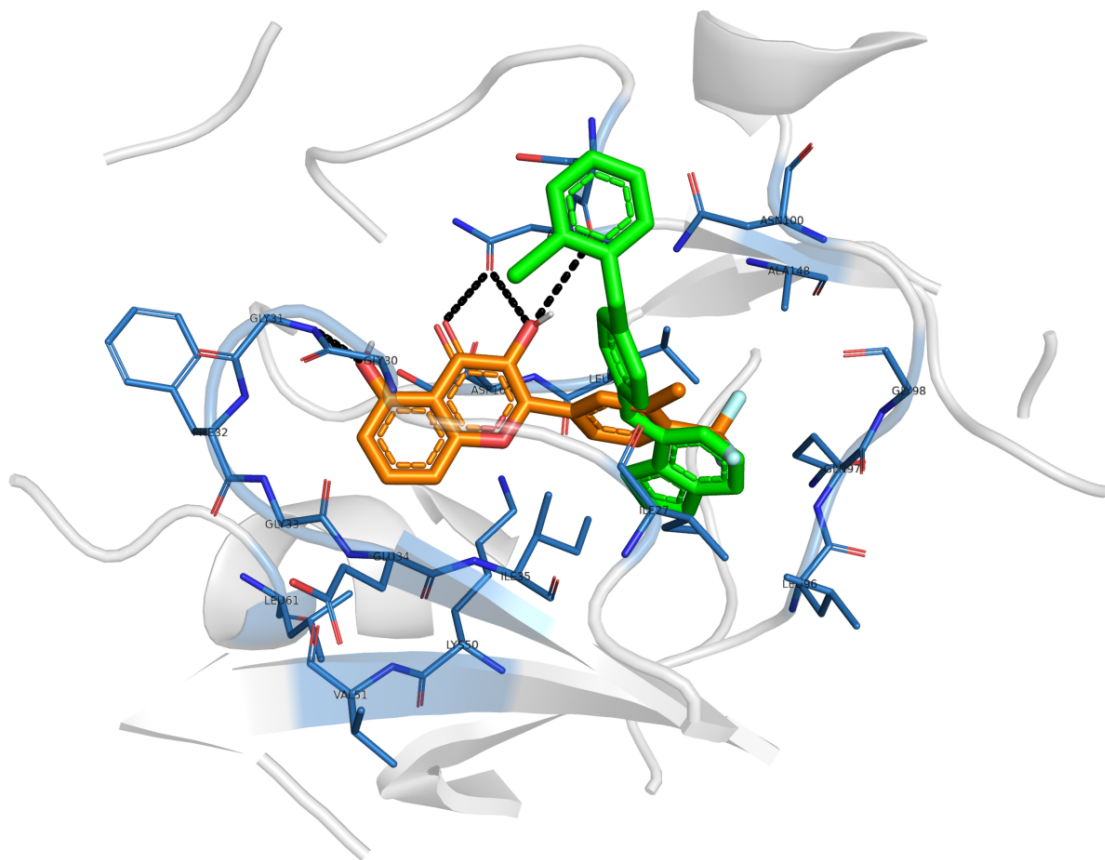
Residues within 4 Å: LYS63 GLU77 LEU81 MET107 MET174 LEU175 ASP176 PHE177

Hydrogen-bond candidates (heavy-atom N/O...N/O ≤ 3.3 Å): none

Distance to removed crystal heteroatoms (raw-pose): NA: closest 9.62 Å; waters: 3 ligand atoms within 3.5 Å.

cand_003 in TTBK2 (7Q8Y)

Vina score **-9.87 (seed 22, mode 1) kcal/mol**. Orange: cand_003. Blue: receptor residues within 4 Å. Green: crystal ligand (removed before docking). Purple: crystal FAD (removed before docking). Black dashes: N/O...N/O pairs within 3.3 Å.



Residues within 4 Å: ILE27 GLY30 GLY31 PHE32 GLY33 ILE35 LYS50 LEU96 GLN97 GLY98 SER145 ASN146 ALA148 LEU162 ASP163

Hydrogen-bond candidates (heavy-atom N/O...N/O ≤ 3.3 Å): GLY31:N ... lig O12 (2.88 Å); SER145:O ... lig O17 (3.29 Å); ASN146:OD1 ... lig O15 (3.21 Å); ASN146:OD1 ... lig O17 (3.02 Å)

Distance to removed crystal heteroatoms (raw-pose): PO4: closest 13.71 Å; waters: 21 ligand atoms within 3.5 Å.

All modes, best seed per receptor

Mode	2V5Z (seed 11)	2Z5X (seed 33)	7JXX (seed 11)	7Q8Y (seed 22)
1	-11.15	-11.44	-8.49	-9.87
2	-11.02	-10.94	-8.19	-9.82
3	-10.57	-9.98	-8.15	-9.65
4	-10.35	-9.64	-8.02	-9.06
5	-10.33	-9.55	-7.95	-9.02
6	-10.08	-9.41	-7.92	-8.87
7	-9.87	-8.77	-7.86	-8.53
8	-9.65	-8.26	-7.85	-8.50
9	-9.31	-7.52	-7.71	-8.34

Full 108-row table (4 receptors x 3 seeds x 9 modes): all_modes_scores.csv. Overlap analysis: overlap_with_removed_atoms.json and mao_a_fad_clash_scan.py in the same folder.